



Yakeen Plus (2025)

PRACTICE TEST - 01

DURATION : 200 Minutes

DATE : 15/09/2024

M. MARKS : 720

Topics Covered

Physics :	Basic Maths & Calculus (Mathematical Tools), Vectors, Units and Measurements
Chemistry :	Some Basic Concept of Chemistry, Redox Reaction
Biology :	(Botany): Cell - The Unit of Life (Zoology): Structural Organization in Animals

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

SECTION-(I) PHYSICS

SECTION - A

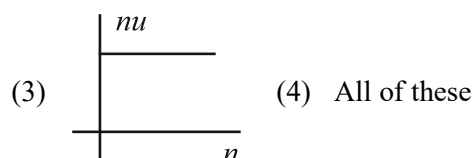
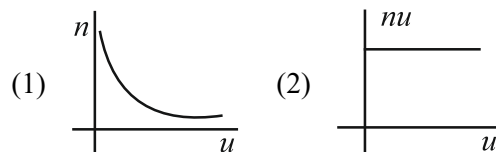
1. Write the dimensions of $\frac{a}{b}$ in the relation

$$P = \frac{a - t^2}{bx}; \text{ where } P \text{ is the pressure, } x \text{ is the}$$

distance, and t is the time:

- (1) $[M^{-1}L^0T^{-2}]$ (2) $[ML^0T^{-2}]$
 (3) $[ML^0T^2]$ (4) $[MLT^{-2}]$
2. If area (A), velocity (v) and density (ρ) are base units, then the dimensional formula of force can be represented:
- (1) $[Av\rho]$ (2) $[Av^2\rho]$
 (3) $[Av\rho^2]$ (4) $[A^2v\rho]$
3. The velocity v of a particle at time t is given by $v = at + (b/(t+c))$, where a , b and c are constants. The dimensions of a , b and c are:
- (1) $[L]$, $[LT]$ and $[LT^{-2}]$
 (2) $[LT^{-2}]$, $[L]$ and $[T]$
 (3) $[L^2]$, $[T]$ and $[LT^2]$
 (4) $[LT^2]$, $[LT]$ and $[L]$
4. A physical quantity P is given by the relation $P = P_0 e^{(-\alpha t^2)}$. If t denotes the time, the dimensions of constant α are:
- (1) $[T]$ (2) $[T^2]$
 (3) $[T^{-1}]$ (4) $[T^{-2}]$
5. The density of a material in CGS system of units is 4 g/cm^3 . In a system of units in which unit of length is 10 cm and unit of mass is 100 g , the value of density of material will be:
- (1) 400 (2) 0.04
 (3) 0.4 (4) 40
6. Which of the following combinations of three dimensionally different physical quantities P , Q , R can never be a meaningful quantity?
- (1) $PQ - R$ (2) $\frac{PQ}{R}$
 (3) $\frac{P - Q}{R}$ (4) $\frac{PR - Q^2}{QR}$

7. Which of the following graph is correct:
 n = magnitude of measurement and u = unit of measurement.



8. Choose the wrong statement:
- (1) a dimensionally correct equation may be correct.
 (2) a dimensionally incorrect equation must be incorrect.
 (3) a dimensionally correct equation may be incorrect.
 (4) a dimensionally incorrect equation may be correct.
9. Vander Waal's gas equation is:
 $\left(P + \frac{a}{V^2}\right)(V-b) = RT$. The dimensions of constant a as given above are:
- (1) $[ML^4T^{-2}]$ (2) $[ML^5T^{-2}]$
 (3) $[ML^3T^{-2}]$ (4) $[ML^2T^{-2}]$
10. If a particle moves from point $P(2, 3, 5)$ to point $Q(3, 4, 5)$. Its displacement vector be:
- (1) $\hat{i} + \hat{j} + 10\hat{k}$ (2) $\hat{i} + \hat{j} + 5\hat{k}$
 (3) $\hat{i} + \hat{j}$ (4) $2\hat{i} + 4\hat{j} + 6\hat{k}$
11. The angles which a vector $\hat{i} + \hat{j} + \sqrt{2}\hat{k}$ make with X , Y and Z axes respectively are:
- (1) $60^\circ, 60^\circ, 60^\circ$
 (2) $45^\circ, 45^\circ, 45^\circ$
 (3) $60^\circ, 60^\circ, 45^\circ$
 (4) $45^\circ, 45^\circ, 60^\circ$
12. If a unit vector is represented by $0.3\hat{i} - 0.4\hat{j} + c\hat{k}$, then the value of ' c ' is:
- (1) $\sqrt{0.75}$ (2) $\sqrt{0.25}$
 (3) $\sqrt{0.01}$ (4) $\sqrt{0.39}$

13. The angle between the vectors $\vec{A}$ and $\vec{B}$ is θ . The value of the triple product $\vec{A} \cdot (\vec{B} \times \vec{A})$ is:

- (1) A^2B (2) Zero
(3) $A^2B \sin \theta$ (4) $A^2B \cos \theta$

14. Given that, $\sin A = \frac{1}{2}$ and $\cos B = \frac{1}{\sqrt{2}}$ then value of $(A + B)$ will be:

- (1) 30° (2) 45°
(3) 75° (4) 15°

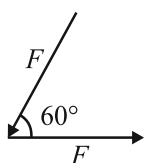
15. If $|\vec{A} \times \vec{B}| = \sqrt{3}\vec{A} \cdot \vec{B}$, then the value of $|\vec{A} + \vec{B}|$ is:

- (1) $\left(A^2 + B^2 + \frac{AB}{\sqrt{3}}\right)^{1/2}$
(2) $A + B$
(3) $\left(A^2 + B^2 + \sqrt{3}AB\right)^{1/2}$
(4) $\left(A^2 + B^2 + AB\right)^{1/2}$

16. If the magnitudes of vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ are 12, 5, and 13 units respectively and $\vec{A} + \vec{B} = \vec{C}$, the angle between vectors A and B is:

- (1) 0 (2) π
(3) $\frac{\pi}{2}$ (4) $\frac{\pi}{4}$

17. Two forces, each numerically equal to 5 N, are acting as shown in the figure. Then the resultant is:



- (1) 2.5 N (2) 5 N
(3) $5\sqrt{3}$ N (4) 10 N

18. If the sum of two unit vectors is a unit vector, then magnitude of difference is:

- (1) $\sqrt{2}$ (2) $\sqrt{3}$
(3) $\frac{1}{\sqrt{2}}$ (4) $\sqrt{5}$

19. Given $x^2 + 7x + 12 = 0$, find the root of x :

- (1) $x = \frac{3}{2}, -4$ (2) $x = -3, -4$
(3) $x = \frac{3}{2}, 4$ (4) $x = \frac{3}{2}, -2$

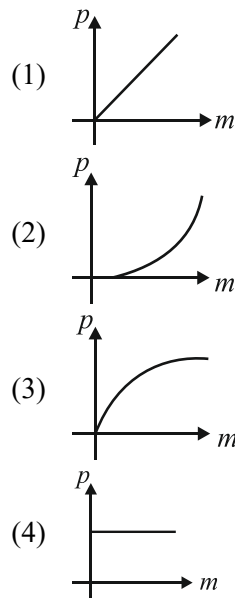
20. The sum of the series $1 + 1/4 + 1/16 + 1/64 + \dots$ ∞ is:

- (1) $8/7$ (2) $6/5$
(3) $5/4$ (4) $4/3$

21. Find the value of $\log_{10} 10^{35}$

- (1) 28 (2) 32
(3) 36 (4) 35

22. Draw graph between momentum and mass of the object. Where E is kinetic energy and it is constant. [$p = \sqrt{2mE}$]



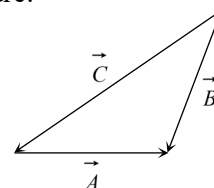
23. $\int_0^1 (x^3 + 1) dx$

- (1) $\frac{1}{4}$ (2) $\frac{3}{4}$
(3) $\frac{5}{4}$ (4) $\frac{7}{4}$

24. The sum of the magnitudes of two vectors is 18 and the magnitude of their resultant is 12. If the resultant is perpendicular to one of the vectors, then what are the magnitudes of the two vectors?

- (1) 5, 13 (2) 6, 12
(3) 7, 11 (4) 8, 11

25. For the figure:



- (1) $\vec{A} + \vec{B} = \vec{C}$
(2) $\vec{B} + \vec{C} = \vec{A}$
(3) $\vec{C} + \vec{A} = \vec{B}$
(4) $\vec{A} + \vec{B} + \vec{C} = \vec{0}$

26. The force F on a sphere of radius ' a ' moving in a medium with velocity ' v ' is given by $F = 6\pi\eta av$. The dimensions of η are:
- (1) $[ML^{-1}T^{-1}]$ (2) $[MT^{-1}]$
 (3) $[MLT^{-2}]$ (4) $[ML^{-3}]$
27. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: When the units of measurement of a quantity are changed, its numerical value changes.
Reason R: Smaller the unit of measurement, smaller is its numerical value.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.
28. If $\vec{D} = \vec{A} - \vec{B}$, and $D^2 = A^2 + B^2$ then find angle between $\vec{A}$ and $\vec{B}$.
- (1) 60° (2) 180°
 (3) 45° (4) 90°
29. Which of the following pair of two vector can't give resultant 3.
- (1) 3, 3 (2) 3, 7
 (3) 2, 5 (4) 2, 1
30. Find differentiation of $y = x^2 - 2x + \sin x$:
- (1) $2x - 2 + \cos x$ (2) $2x + 2 + \sin x$
 (3) $2x - 2 + \sin x$ (4) $x^2 - 2x$

31. Match List-I with List-II.

List-I		List-II	
(A)	Surface tension	(I)	$\text{Kg m}^{-1} \text{s}^{-1}$
(B)	Pressure	(II)	Kg m s^{-1}
(C)	Viscosity	(III)	$\text{Kg m}^{-1} \text{s}^{-2}$
(D)	Impulse	(IV)	Kg s^{-2}

Choose the **correct** answer from the option given below:

- (1) A-IV, B-III, C-II, D-I
 (2) A-IV, B-III, C-I, D-II
 (3) A-III, B-IV, C-I, D-II
 (4) A-II, B-I, C-III, D-IV
32. The pair of quantities having same dimensions is;
- (1) impulse and surface tension.
 (2) angular momentum and work.
 (3) work and torque.
 (4) young's modulus and energy.

33. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Astronomical unit, light year and parsec measures distance.

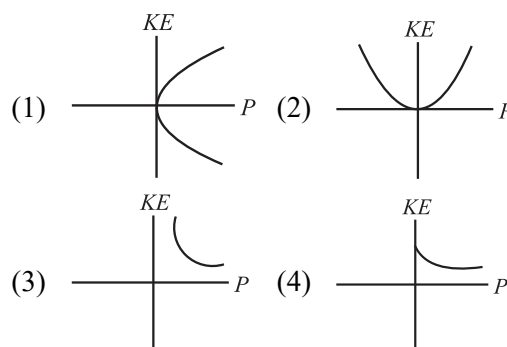
Reason R: Each has different dimension of distance.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.
34. $(1+x)^3$ find the value, if $x \ll 1$.
- (1) $1+x$
 (2) $1-3x$
 (3) $1+3x$
 (4) $1+3x+3x^2+x^3$
35. If $a, a+d, a+2d, \dots, a+(n-1)d$ here a = first term, d = common difference, then sum of n terms S_n is:
- (1) $\frac{n}{2}[2a+(n-1)d]$
 (2) $\frac{n}{2}[1^{\text{st}} \text{ term} + n^{\text{th}} \text{ term}]$
 (3) Option (1) and (2)
 (4) $\frac{n}{2}[2a+(n+1)d]$

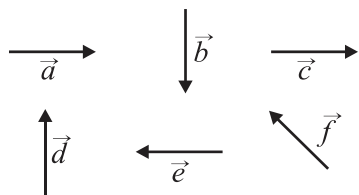
SECTION - B

36. If $KE = \frac{P^2}{2m}$, then draw graph between KE and P .



37. If $y = \sin 4t$, then value of dy/dt is:
- (1) $4 \sin 4t$
 (2) $\cos 4t/4$
 (3) $4 \cos 4t$
 (4) $-4 \cos 4t$

38. Six vectors $\vec{a}$ to $\vec{f}$ have the magnitudes and directions indicated in the figure. Which of the following statements is true?



- (1) $\vec{a} + \vec{c} = \vec{f}$ (2) $\vec{d} + \vec{c} = \vec{f}$
 (3) $\vec{d} + \vec{e} = \vec{f}$ (4) $\vec{b} + \vec{e} = \vec{f}$

39. If a vector $2\hat{i} + 3\hat{j} + 8\hat{k}$ is perpendicular to the vector $4\hat{i} - 4\hat{j} + \alpha\hat{k}$, then the value of α is:

- (1) -1 (2) $\frac{1}{2}$
 (3) $-\frac{1}{2}$ (4) 1

40. If $|\vec{A} + \vec{B}| = |\vec{A}| = |\vec{B}|$ then angle between A and B will be:

- (1) 90° (2) 120°
 (3) 0° (4) 60°

41. For a body, angular velocity $\vec{\omega} = \hat{i} - \hat{j} + \hat{k}$ and position vector $\vec{r} = \hat{i} + \hat{j} + \hat{k}$, then its velocity ($\vec{v} = \vec{\omega} \times \vec{r}$) is:

- (1) $-5\hat{i} + 2\hat{j} + 3\hat{k}$ (2) $-5\hat{i} + 2\hat{j} - 3\hat{k}$
 (3) $-2\hat{i} + 2\hat{k}$ (4) $-5\hat{i} - 2\hat{j} - 3\hat{k}$

42. If the vectors $(\hat{i} + \hat{j} + \hat{k})$ and $\hat{i}$ form two sides of a triangle, the area of the triangle is:

- (1) $\frac{1}{\sqrt{2}}$ (2) $2\sqrt{3}$
 (3) $\frac{3}{\sqrt{2}}$ (4) $3\sqrt{2}$

43. $\alpha = \frac{F}{v^2} \sin(\beta t)$ (where v = velocity,

F = force, t = time)

Find the dimension of α and β respectively

- (1) $M^1 L^{-1} T^0, M^1 L^{-1} T^0$
 (2) $M^1 L^1 T^{-2}, M^1 L^{-1} T^0$
 (3) $M^{-1} L^{-1} T^{-2}, M^1 L^1 T^{-2}$
 (4) $M^1 L^{-1} T^0, M^0 L^0 T^{-1}$

44. If a vector $\vec{P}$ making angles α, β and γ respectively with X, Y and Z axes respectively.

Then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma =$

- (1) 0 (2) 1
 (3) 2 (4) 3

45. Forces 7 N, 24 N, 25 N act at a point in mutually perpendicular directions. The magnitude of the resultant force is :

- (1) 19 N (2) 13 N
 (3) 26 N (4) $25\sqrt{2}$ N

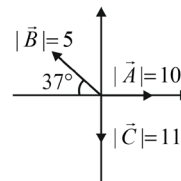
46. A dimensionless quantity:

- (1) Never has a unit (2) Always has a unit
 (3) May have a unit (4) Does not exist

47. A unitless quantity

- (1) Does not exist
 (2) Always has a nonzero dimension
 (3) Never has a nonzero dimension
 (4) May have a nonzero dimension

48. Find the resultant of following vectors:



- (1) 8 (2) 6
 (3) 10 (4) 20

49. The frequency of vibration f of a mass m suspended from a spring of spring constant K is given by a relation of this type $f = C m^x K^y$; where C is a dimensionless quantity. The value of x and y are:

- (1) $x = \frac{1}{2}, y = \frac{1}{2}$ (2) $x = -\frac{1}{2}, y = -\frac{1}{2}$
 (3) $x = \frac{1}{2}, y = -\frac{1}{2}$ (4) $x = -\frac{1}{2}, y = \frac{1}{2}$

50. The velocity of water waves v may depend upon their wavelength λ , the density of water ρ and the acceleration due to gravity g . The method of dimensions gives the relation between these quantities as:

- (1) $v^2 \propto g$ (2) $v^2 \propto g\lambda\rho$
 (3) $v^2 \propto g\lambda$ (4) $v^2 \propto g^{-1}\lambda^{-3}$

SECTION-(II) CHEMISTRY

SECTION - A

51. The number of neutrons in 5 g of D_2O (D is 2_1H) are:
 (1) $0.25 N_A$ (2) $2.5 N_A$
 (3) $1.1 N_A$ (4) $25 N_A$
52. If Avogadro number N_A , is changed from $6.022 \times 10^{23} \text{ mol}^{-1}$ to $6.022 \times 10^{20} \text{ mol}^{-1}$, this would change:
 (1) the ratio of chemical species to each other in a balanced equation.
 (2) the ratio of elements to each other in a compound.
 (3) the definition of mass in units of grams.
 (4) the mass of one mole of carbon.
53. Which of the following statements is/are **true**?
 (1) An element of a substance contains only one kind of atoms.
 (2) A compound can be decomposed into its components.
 (3) All homogeneous mixtures are called as solutions.
 (4) All of these
54. The density of water is 1 g/mL . What is the volume occupied by 1 molecule of water?
 ($1 \text{ amu} = 1.66 \times 10^{-24} \text{ g}$)
 (1) $1.44 \times 10^{-23} \text{ mL}$
 (2) 1 mL
 (3) 18 mL
 (4) $2.98 \times 10^{-23} \text{ mL}$
55. Average atomic mass of magnesium is 24.31 amu . This magnesium is composed of 79 mole % of ${}^{24}\text{Mg}$ and remaining 21 mole % of ${}^{25}\text{Mg}$ and ${}^{26}\text{Mg}$. Calculate mole % of ${}^{26}\text{Mg}$.
 (1) 10% (2) 11%
 (3) 15% (4) 16%
56. If water sample are taken from sea, rivers or lake, they will be found to contain hydrogen and oxygen in the approximate ratio of 1 : 8. This indicates the law of:
 (1) multiple proportion.
 (2) definite proportion.
 (3) conservation of mass.
 (4) None of these
57. The modern atomic weight scale is based on:
 (1) C^{12} isotope. (2) O^{16} isotope.
 (3) H^1 isotope. (4) C^{13} isotope.
58. The law of multiple proportion is illustrated by the two compounds:
 (1) sodium chloride and sodium bromide.
 (2) ordinary water and heavy water.
 (3) caustic soda and caustic potash.
 (4) sulphur dioxide and sulphur trioxide.
59. What percentage of oxygen is present in the compound $CaCO_3 \cdot 3Ca_3(PO_4)_2$? (Molar mass of $CaCO_3 = 100 \text{ g/mol}$, Molar mass of $Ca_3(PO_4)_2 = 310 \text{ g/mol}$)
 (1) 23.3%
 (2) 45.36%
 (3) 41.94%
 (4) 17.08%
60. Two element X (atomic mass = 75) and Y (atomic mass = 16) combine to give a compound having 75.8% of X. The formula of the compound is:
 (1) XY (b) X_2Y
 (3) X_2Y_2 (4) X_2Y_3
61. 1 amu is equal to:
 (1) $\frac{1}{12}$ mass of C-12 isotope.
 (2) $\frac{1}{14}$ mass of O-16 isotope.
 (3) 1 g of H_2 .
 (4) $1.66 \times 10^{-24} \text{ kg}$.
62. The relation between molarity (M) and molality (m) is given by: (ρ = density of solution (g/mL), M_1 = molecular weight of solute)
 (1) $m = \frac{1000M}{1000\rho - M_1}$
 (2) $m = \frac{1000\rho M}{1000\rho - MM_1}$
 (3) $m = \frac{1000MM_1}{1000\rho - MM_1}$
 (4) $m = \frac{1000M}{1000\rho - MM_1}$

63. The oxidation state(s) of Cl in CaOCl_2 (bleaching powder) is/are:
 (1) +1 only
 (2) -1 only
 (3) +1 and -1
 (4) +2 and -2
64. Molecular mass of tribasic acid is W. Its equivalent weight will be:
 (1) $\frac{W}{2}$ (2) $\frac{W}{3}$
 (3) W (4) 3W
65. A redox reaction is:
 (1) exothermic.
 (2) endothermic.
 (3) neither exothermic nor endothermic.
 (4) either exothermic or endothermic.
66. The equivalent weight of H_2SO_4 in the reaction:
 $2\text{KMnO}_4 + 3\text{H}_2\text{SO}_4 + 10\text{HCl} \rightarrow 2\text{MnSO}_4 + \text{K}_2\text{SO}_4 + 5\text{Cl}_2 + 8\text{H}_2\text{O}$, is:
 (1) $\frac{M}{2}$ (2) M
 (3) $\frac{3M}{10}$ (4) $\frac{3M}{5}$
67. Which of these substance is a good reducing agent?
 (1) HI
 (2) KBr
 (3) FeCl_3
 (4) KClO_3
68. When potassium permanganate is titrated against ferrous ammonium sulphate in acidic medium, the equivalent weight of potassium permanganate is:
 $\text{MnO}_4^- + 8\text{H}^+ + 5\text{Fe}^{2+} \rightarrow 5\text{Fe}^{3+} + \text{Mn}^{2+} + 4\text{H}_2\text{O}$.
 (1) $\frac{\text{Molecular mass}}{3}$
 (2) $\frac{\text{Molecular mass}}{5}$
 (3) $\frac{\text{Molecular mass}}{2}$
 (4) $\frac{\text{Molecular mass}}{10}$
69. The volume of 1.0 g of hydrogen at NTP is:
 (1) 2.24 L
 (2) 22.4 L
 (3) 1.12 L
 (4) 11.2 L
70. 11 g of a gas occupy 5.6 litres of volume at STP. The gas is:
 (1) NO
 (2) N_2O_4
 (3) CO
 (4) CO_2
71. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: 22.4 L of He gas is having N_A atoms of He at 1 atm and 273 K.
Reason R: 22.4 L of any gas at 1 atm and 273 K is having N_A atoms.
 In the light of the above statements, choose the **correct** answer from the options given below:
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.
72. Given below are two statements:
Statement I: Atoms can neither be created nor destroyed.
Statement II: Under similar condition of temperature and pressure, equal volume of gases does not contain equal number of atoms.
 In the light of the above statements, choose the *most appropriate* answer from the options given below:
 (1) Statement I is correct but Statement II is incorrect.
 (2) Statement I is incorrect but Statement II is correct.
 (3) Both Statement I and Statement II are correct.
 (4) Both Statement I and Statement II are incorrect.

73. The oxidation number of sulphur in $\text{H}_2\text{S}_2\text{O}_7$ and iron in $\text{K}_4[\text{Fe}(\text{CN})_6]$ is respectively:

- (1) +6 and +2
- (2) +2 and +2
- (3) +8 and +2
- (4) +6 and +4

74. Molarity of liquid HCl with density equal to 1.17 g/cc is: (Molar mass of $\text{HCl} = 36.5 \text{ g/mol}$)

- (1) 36.5 mol L^{-1} (2) 18.25 mol L^{-1}
- (3) 32.05 mol L^{-1} (4) 4.65 mol L^{-1}

75. The total number of protons in 10 g of calcium carbonate is: ($N_A = 6.022 \times 10^{23}$) (Molar mass of $\text{CaCO}_3 = 100 \text{ g/mol}$)

- (1) 1.505×10^{24}
- (2) 2.047×10^{24}
- (3) 3.011×10^{24}
- (4) 4.095×10^{24}

76. The total number of electrons in 18 mL of water (density = 1 g mL^{-1}) is:

- (1) 6.022×10^{23}
- (2) 6.022×10^{25}
- (3) 6.022×10^{24}
- (4) $6.022 \times 18 \times 10^{23}$

77. Molecular mass of dry air is:

- (1) less than moist air.
- (2) greater than moist air.
- (3) equal to moist air.
- (4) may be greater or less than moist air.

78. For $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, which is the **correct** mole relationship?

- (1) $9 \times \text{Moles of Cu} = \text{Moles of O}$
- (2) $5 \times \text{Moles of Cu} = \text{Moles of O}$
- (3) $9 \times \text{Moles of Cu} = \text{Moles of O}_2$
- (4) $\text{Moles of Cu} = 5 \times \text{Moles of O}$

79. Boron has two isotopes B^{10} and B^{11} whose percentage abundances are 19.6% and 80.4% respectively. What is the average atomic mass of boron?

- (1) 10.8 amu (2) 9.8 amu
- (3) 7.7 amu (4) 8.8 amu

80. Calculate the gram atoms in 2.3 g of sodium. (Molar mass of $\text{Na} = 23 \text{ g/mol}$)

- (1) 0.1 (2) 1.3
- (3) 1 (4) 2

81. A piece of copper weighs 0.635 g . How many atoms of copper does it contain? (Molar mass of $\text{Cu} = 63.5 \text{ g/mol}$)

- (1) 6.022×10^{21} (2) 60.22×10^{21}
- (3) 6.022×10^{23} (4) 602.2×10^{23}

82. 3 g of hydrocarbon on combustion with 11.2 g of oxygen produce 8.8 g of CO_2 and 5.4 g of H_2O . The data illustrate the law of:

- (1) conservation of mass.
- (2) multiple proportions.
- (3) constant proportions.
- (4) avogadro's law.

83. Insulin contains 3.4% sulphur. Then the minimum molecular mass of the insulin is about: (Molar mass of sulphur = 32 g/mol)

- (1) 940 amu (2) 9400 amu
- (3) 3600 amu (4) 970 amu

84. An unknown element forms an oxide. What will be the equivalent mass of the element if the oxygen content is 20% by mass:

- (1) 16 (2) 32
- (3) 8 (4) 64

85. The vapour densities of two gases are in the ratio of $1:3$. Their molecular masses are in the ratio of:

- (1) $1:3$ (2) $1:2$
- (3) $2:3$ (4) $3:1$

SECTION - B

86. The atomic composition of the entire universe is approximately given in the table below:

Atom	% of total number of atoms
H	93
He	7

Hydrogen atoms constitute what percentage of the universe by mass?

- (1) 77% (2) 23%
- (3) 37% (4) 73%

87. Match List-I with List-II.

List-I (Quantity)		List-II (Units)	
(A)	Density	(I)	bar
(B)	Pressure	(II)	kg m ³
(C)	Length	(III)	eV
(D)	Energy	(IV)	A°

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
- (2) A-IV, B-III, C-II, D-I
- (3) A-II, B-III, C-IV, D-I
- (4) A-II, B-I, C-III, D-IV

88. An organic compound containing C and H has 92.30% carbon. Its empirical formula is:

- (1) CH
- (2) CH₃
- (3) CH₂
- (4) CH₄

89. Gram atomic mass of oxygen is:

- (1) 16 amu
- (2) 16 g
- (3) 32 amu
- (4) 32 g

90. In a reaction, $A + 5B \rightarrow C + 3D$

Calculate the produced mole of C. (Given A = 10 mole, B = 10 mole)

- (1) 2 mole
- (2) 5 mole
- (3) 6 mole
- (4) 1 mole

91. The number of F⁻ ions in 4.2 g AlF₃ is: (Molar mass of Al = 27 g/mol, Molar mass of F = 19 g/mol)

- (1) 0.05
- (2) 9.033×10^{22}
- (3) 3.011×10^{22}
- (4) 0.15

92. Which order of compounds is according to the decreasing order of the oxidation state of nitrogen?

- (1) HNO₃ > NO > NH₄Cl > N₂
- (2) HNO₃ > NO > N₂ > NH₄Cl
- (3) HNO₃ > NH₄Cl > NO > N₂
- (4) NO > HNO₃ > NH₄Cl > N₂

93. In which of the following reactions, H₂O₂ is acting as a reducing agent?

- (1) SO₂ + H₂O₂ → H₂SO₄
- (2) 2KI + H₂O₂ → 2KOH + I₂
- (3) PbS + 4H₂O₂ → PbSO₄ + 4H₂O
- (4) Ag₂O + H₂O₂ → 2Ag + H₂O + O₂

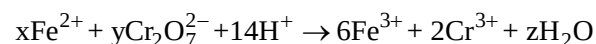
94. The oxidation state of chromium is +6 in:

- (1) K₃CrO₈
- (2) Cr₂O₃
- (3) Cr₂(SO₄)₃
- (4) CrO₅

95. Which of the following is a redox-reaction?

- (1) $2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Zn} \rightarrow \text{Na}_2[\text{Zn}(\text{CN})_4] + 2\text{Ag}$
- (2) $\text{BaO}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + \text{H}_2\text{O}_2$
- (3) $\text{N}_2\text{O}_5 + \text{H}_2\text{O} \rightarrow 2\text{HNO}_3$
- (4) $\text{AgNO}_3 + \text{KI} \rightarrow \text{AgI} + \text{KNO}_3$

96. In following redox reaction,



The values of x, y, z are:

- (1) x = 6, y = 1, z = 7
- (2) x = 3, y = 6, z = 8
- (3) x = 6, y = 2, z = 7
- (4) x = 4, y = 6, z = 7

97. Which of the following ion **cannot** act as an oxidizing agent?

- (1) MnO₄⁻
- (2) CrO₄²⁻
- (3) I⁻
- (4) Fe³⁺

98. The pair of compounds having metals in their highest oxidation state is:

- (1) MnO₂, FeCl₃
- (2) [MnO₄]⁻, CrO₂Cl₂
- (3) [Fe(CN)₆]³⁻, [Co(CN)₃]
- (4) [NiCl₄]²⁻, [CoCl₄]⁻

99. The sum of oxidation states of all carbon atoms in toluene molecule is:

- (1) -1
- (2) -7/8
- (3) -8/7
- (4) -8

100. Which of the following statements is **true** about oxidation state of S in Na₂S₄O₆?

- (1) All S-atoms are in +2.5 oxidation state.
- (2) All S-atoms are in +2 oxidation state.
- (3) Two S-atoms are in 0 oxidation state and other two is in +5 oxidation state.
- (4) Two S-atoms are in -1 oxidation state and other two is in +6 oxidation state.

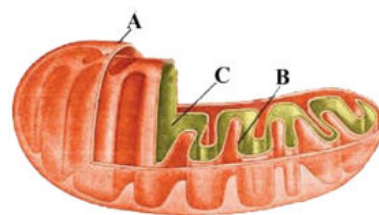
SECTION-(III) BOTANY

SECTION - A

- 101.** A student compared onion cell & human cheek cell then cell of onion peel differs from a cell of human cheek in:
- (1) the type of outer boundary.
 - (2) having both 70S and 80S ribosomes.
 - (3) having two non-membrane bound cell organelles.
 - (4) double membrane bound nucleus.
- 102.** Read the given statements.
 P: It can occupy upto 90% volume of a plant cell.
 Q: It is bound by a single membrane called tonoplast.
 Identify the cell organelle on the basis of above features.
- (1) Lysosome
 - (2) Vacuole
 - (3) Ribosomes
 - (4) Peroxisome
- 103.** Choose the **odd** one w.r.t. inclusion bodies of bacterial cell.
- (1) Gas vacuoles
 - (2) Phosphate granules
 - (3) Glycogen granules
 - (4) Extrachromosomal DNA
- 104.** Plasmid of bacteria **cannot** be:
- (1) larger than genomic DNA.
 - (2) circular.
 - (3) found outside the genomic DNA.
 - (4) used to monitor bacterial transformation.
- 105.** In mammals, proteinaceous structure which maintains the shape of a cell, is:
- (1) cell wall.
 - (2) cytoplasm.
 - (3) cytoskeleton.
 - (4) vacuole.
- 106.** One **cannot** found in power house of the cell?
- (1) Circular DNA
 - (2) Site of anaerobic respiration
 - (3) Infoldings of its inner membrane
 - (4) RNA molecules
- 107.** Cell membrane of animal cell:
- (1) does not contain cholesterol.
 - (2) contains lipids and proteins only.
 - (3) contains lipids, proteins and carbohydrates.
 - (4) does not contain carbohydrates.
- 108.** The nuclear pores are the passages through which movement of RNA and protein molecules takes place in:
- (1) one direction i.e. from the nucleus to the cytoplasm only.
 - (2) both directions between the nucleus and cytoplasm.
 - (3) inside of the nucleus only.
 - (4) uncontrolled manner.
- 109.** Sometimes few chromosomes have non-staining secondary constriction at a constant location. This gives the appearance of a small fragment called ____.
- (1) short arm
 - (2) satellite
 - (3) kinetochore
 - (4) terminal centromere
- 110.** In some prokaryotes like cyanobacteria, there are membranous extensions into the cytoplasm which contain pigments are called as:
- (1) mesosomes.
 - (2) chromatophores.
 - (3) microbodies.
 - (4) polysomes.
- 111.** The site for active ribosomal RNA synthesis in human is:
- (1) ribosome.
 - (2) endoplasmic reticulum.
 - (3) Golgi apparatus.
 - (4) nucleolus.
- 112.** Which of the following is found in animal cells?
- (1) Microbodies
 - (2) Cell wall
 - (3) Chromoplast
 - (4) Leucoplast
- 113.** In metacentric chromosome, the centromere is located:
- (1) at the middle of chromosome.
 - (2) slightly away from the middle of chromosome.
 - (3) close to one end of chromosome.
 - (4) at the end of the chromosome.

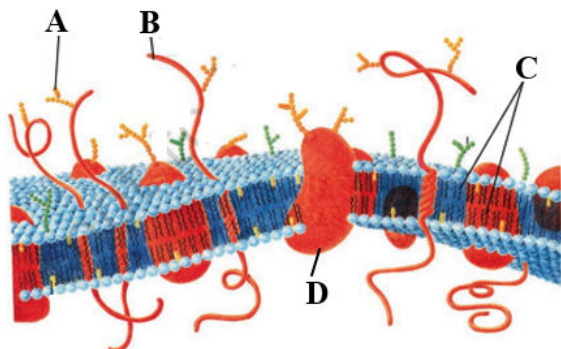
- 114.** Mesosome found in gram positive bacteria is:
- (1) small circular DNA outside the genomic DNA.
 - (2) a specialised differentiated form of cell membrane.
 - (3) fluid matrix of the cytoplasm.
 - (4) a structure on their cell wall.
- 115.** The core of eukaryotic cilia is called axoneme, which is made up of:
- (1) filament, hook and basal body.
 - (2) 9 centrally located doublets and 2 microtubules at periphery.
 - (3) 2 centrally located microtubules and 9 doublets at periphery.
 - (4) central hub and 11 peripheral doublets.
- 116.** Which of the following is **not** feature of cell wall?
- (1) Protects the cell from bursting.
 - (2) Provides barrier to undesirable macromolecules.
 - (3) Protect the cells from infections.
 - (4) Involves in active transport of materials.
- 117.** The lysosomal hydrolases are able to digest:
- (1) proteins and carbohydrates only.
 - (2) proteins, carbohydrates, lipids and nucleic acids.
 - (3) proteins and lipids only.
 - (4) carbohydrates, lipids and nucleic acid only.
- 118.** Select the **incorrect** statement.
- (1) Egg of ostrich is the largest isolated single cell.
 - (2) Ribosomes is membrane less organelle.
 - (3) The shape of the cells may vary with the function they perform.
 - (4) Genetic material is present in well defined nucleus in both prokaryotes and eukaryotes.
- 119.** Proteins formed by rough endoplasmic reticulum modified in Golgi apparatus is going to release from its:
- (1) concave *cis* face.
 - (2) convex *cis* face.
 - (3) concave *trans* face.
 - (4) convex *trans* face.
- 120.** Plastids that gives distinct colours (yellow, orange or red colour) to the plants are known as:
- (1) chloroplasts.
 - (2) chromoplasts.
 - (3) elaioplasts.
 - (4) aleuoplasts.

- 121.** Which of the following mature cells lack nucleus?
- (1) Erythrocytes
 - (2) White blood cells
 - (3) Human cheek cells
 - (4) Onion cells
- 122.** Identify the **mismatched** pair.
- (1) Cristae – Mitochondria
 - (2) Na^+/K^+ Pump– Active transport
 - (3) Thylakoids – Lysosomes
 - (4) Microtubules – Cytoskeleton
- 123.** Identify the **correct** statement w.r.t kinetochore.
- (1) Terminal ends of chromosomes.
 - (2) Present around primary constriction and help in formation of chromatids.
 - (3) Present near nuclear envelope and take part in spindle formation.
 - (4) Disc shaped structure present on the sides of centromere.
- 124.** Identify from the following which is **not** considered as a part of the endomembrane system.
- (1) Lysosome
 - (2) Peroxisome
 - (3) Vacuole
 - (4) Golgi complex
- 125.** 80S ribosomes have the subunits:
- (1) 60S and 40S.
 - (2) 55S and 25S.
 - (3) 50S and 30S.
 - (4) 16S and 23S.
- 126.** Identify the labels **A, B & C** in the given diagram and choose the option that has their **correct** features.



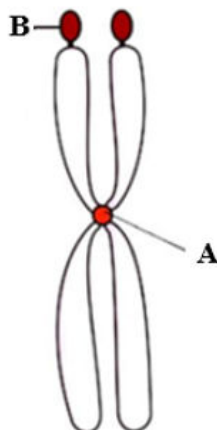
	A	B	C
(1)	Forms a number of infoldings	Increases surface area	Contains 70S ribosomes
(2)	Forms the continuous limiting boundary of the organelle.	Has small circular dsDNA	Increases surface area
(3)	Filled with a dense homogeneous substance.	Crista	Lack dsDNA
(4)	Forms the continuous limiting boundary of the organelle.	Increases surface area	Contains 70S ribosomes

127. Consider the given diagram and select the **correct** option w.r.t. labelled components (A-D).



	A	B	C	D
(1)	Lipid	Cholesterol	Peripheral protein	Integral protein
(2)	Cholesterol	Integral protein	Sugar	Peripheral protein
(3)	Sugar	Peripheral protein	Phospholipid bilayer	Integral protein
(4)	Sugar	Integral protein	Cholesterol	Lipid

128. The following diagram represents a structure of chromosome. Identify the structures marked as A and B.



	A	B
(1)	Satellite	Primary constriction
(2)	Satellite	Secondary constriction
(3)	Satellite	Centromere
(4)	Centromere	Satellite

129. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Bacillus	(I)	Spiral shaped
(B)	Coccus	(II)	Comma shaped
(C)	Vibrio	(III)	Spherical shaped
(D)	Spirillum	(IV)	Rod shaped

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
- (2) A-IV, B-III, C-II, D-I
- (3) A-IV, B-III, C-I, D-II
- (4) A-I, B-II, C-III, D-IV

130. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Inclusion bodies	(I)	Increases surface area and enzymatic content of plasma membrane
(B)	Plasmid DNA	(II)	Several ribosomes attach to a single mRNA
(C)	Polysome	(III)	Reserve materials in the cytoplasm of prokaryotes
(D)	Mesosome	(IV)	Confers certain unique phenotypic characters to bacteria

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-IV, C-II, D-I
- (4) A-III, B-II, C-IV, D-I

131. Read the following statements.

- A. Golgi apparatus is the important site of formation of glycoproteins and glycolipids.
- B. Enzymes of lysosomes are optimally active at basic pH.
- C. Majority of the chloroplasts of the green plants are found in the mesophyll cells.
- D. Endoplasmic reticulum is involved in protein synthesis only.

How many is/are **correct**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

132. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Typical bacteria	(I)	1 – 2 μm
(B)	Viruses	(II)	0.1 μm
(C)	PPLO	(III)	0.02 μm
(D)	Typical eukaryotic cell	(IV)	10 – 20 μm

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-II, B-IV, C-I, D-III
- (3) A-I, B-III, C-II, D-IV
- (4) A-IV, B-I, C-II, D-III

133. Match List-I with List-II.

List-I		List-II	
(A)	Anton Von Leeuwenhoek	(I)	Proposed fluid mosaic model of plasma membrane
(B)	Rudolf Virchow	(II)	First described nucleus as a cell organelle
(C)	Robert Brown	(III)	First saw and described a live cell
(D)	Singer and Nicolson	(IV)	Give final shape to the cell theory

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
- (2) A-II, B-IV, C-III, D-I
- (3) A-III, B-I, C-IV, D-II
- (4) A-IV, B-II, C-I, D-III

134. Read the following statements:

- A. Golgi apparatus is non-membrane bound organelle.
- B. The middle lamella of cell wall is a layer mainly made up of calcium pectate.
- C. Cell wall helps in cell-to-cell interaction.
- D. Plastids are easily observed under the microscope as they are large.

How many is/are **incorrect**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

135. Match List-I with List-II.

List-I		List-II	
(A)	Material of the nucleus stained by the basic dyes	(I)	Osmoregulation
(B)	Pericentriolar material	(II)	Chromatin
(C)	Arise from the centriole like structure	(III)	Cilia and flagella
(D)	Contractile vacuoles	(IV)	Centrosome

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
- (2) A-IV, B-II, C-III, D-I
- (3) A-II, B-IV, C-III, D-I
- (4) A-IV, B-I, C-II, D-III

SECTION - B

136. Read the following statements:

- A. The eukaryotic cells have membrane bound nucleus.
- B. All living organisms are composed of cells and products of cells.
- C. All cells arise from pre-existing cells.
- D. Prokaryotic cells are represented by bacteria and mycoplasma.

Which of the above statements are considered to be part of cell theory after Virchow contribution?

- (1) C only
- (2) B and C only
- (3) A and C only
- (4) A, B and D only

137. Identify the **incorrect statement(s).**

- A. All prokaryotes have a cell wall surrounding the cell membrane except in blue green algae.
- B. The eukaryotic cells have membrane bound distinct structures called organelles.
- C. The prokaryotic cell envelope consists of a tightly bound three layered structure.
- D. Prokaryotic flagella is structurally different from eukaryotic flagella.

Choose the most appropriate answer from the options given below:

- (1) A only
- (2) B, C and D only
- (3) A and C only
- (4) A, B and D only

138. Identify the **correct statements.**

- A. The matrix of mitochondria does not contain the components required for the synthesis of proteins.
- B. The centrioles form spindle fibres that give rise to spindle apparatus during cell division in animal cells.
- C. The cisternae of Golgi bodies are concentrically arranged near the nucleus.
- D. Vacuoles contains water, sap, excretory product and other materials not useful for the cell.

Choose the most appropriate answer from the options given below:

- (1) B and D only
- (2) B, C and D only
- (3) A and C only
- (4) A, B and D only

139. Identify the **incorrect** statements.

- A. The mitochondria is visible under the microscope without staining.
- B. Larger and more numerous nucleoli are present in cells actively carrying out respiration.
- C. Food vacuoles are the example of inclusion bodies.
- D. Mitochondria is sausage-shaped or cylindrical structure.

Choose the most appropriate answer from the options given below:

- (1) B and D only
- (2) B, C and D only
- (3) A and C only
- (4) A, B and C only

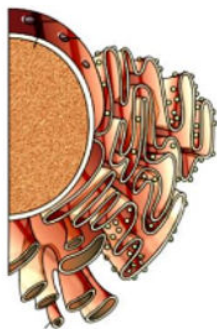
140. Identify the **correct** statements.

- A. Middle lamella holds or glues the different neighbouring plant cells together.
- B. Leucoplasts contain chlorophyll pigments.
- C. Functions of the mitochondria, chloroplast and peroxisomes are coordinated with ER.
- D. The Golgi apparatus in the cell principally performs the function of packaging materials.

Choose the most appropriate answer from the options given below:

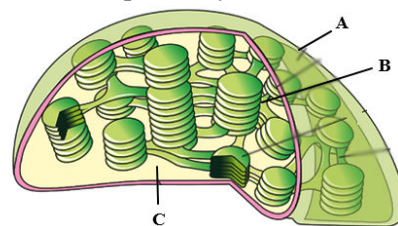
- (1) B and D only
- (2) B, C and D only
- (3) A and D only
- (4) A, B and C only

141. Identify the **incorrect** statement regarding the given figure.



- (1) ER divides the intracellular space into two distinct compartments, luminal and extra luminal compartments.
- (2) The endoplasmic reticulum bearing ribosomes on their surface is called rough endoplasmic reticulum (RER).
- (3) RER is frequently observed in the cells actively involved in protein synthesis and secretion.
- (4) RER are extensive and continuous with the inner membrane of the nucleus.

142. In the diagram of chloroplast given below, identify A, B and C respectively.



	A	B	C
(1)	Inner membrane	Grana	Stroma
(2)	Inner membrane	Stroma	Grana
(3)	Outer membrane	Stroma	Grana
(4)	Outer membrane	Grana	Stroma

143. Read the following statements.

- A. The chromoplasts contain chlorophyll and carotenoid pigments which are responsible for trapping light energy essential for photosynthesis.
- B. In the chromoplasts water soluble carotenoid pigments like carotene, xanthophylls and others are present.
- C. The ribosomes of the chloroplasts are smaller (70S) than the cytoplasmic ribosomes (80S).
- D. Flagella are longer than cilia and responsible for cell movement.
- E. Cilia are small structures which work like oars, causing the movement of either the cell or the surrounding fluid.

How many is/are **correct**?

- (1) One
- (2) Two
- (3) Three
- (4) Five

144. Read the following statements:

- A. The eukaryotes include all the protists, plants, animals and fungi.
- B. The detailed structure of the membrane was studied only after the advent of the electron microscope in the 1980s.
- C. Chemical studies on the cell membrane, especially in human red blood cells (RBCs), enabled the scientists to deduce the possible structure of plasma membrane.
- D. In addition to phospholipids cell membrane also contains cholesterol.

How many is/are **correct**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

145. Read the following statements:

- A. In the chloroplast, thylakoids are arranged in stacks like the piles of coins called grana.
- B. Flagella in bacteria help in motility and capturing foods.
- C. Cisternae in Golgi apparatus are stacked parallel to each other.
- D. Organisation of all prokaryotic cells are fundamentally similar.

How many is/are **correct**?

- (1) One (2) Two
- (3) Three (4) Four

146. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The content of nucleolus is continuous with rest of the nucleoplasm.

Reason R: Nucleolus is not a membrane bound structure.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

147. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The cell wall of a young plant cell, the secondary wall is capable of growth, which gradually diminishes as the cell matures.

Reason R: The secondary wall is formed on the inner (towards membrane) side of the primary cell wall.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

148. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The space limited by the inner membrane of the chloroplast is called the stroma.

Reason R: There are flat membranous tubules in chloroplast called the stroma lamellae.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

149. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Ribosomes are the agranular structures.

Reason R: Ribosomes are composed of ribonucleic acid (RNA) and proteins.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Each chromosome is visible only in dividing cell.

Reason R: A single human cell has approximately two metre long thread DNA.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

SECTION-(IV) ZOOLOGY

SECTION - A

151. Which of the following connective tissue attach bones to bones?
- (1) Areolar tissue
 - (2) Adipose tissue
 - (3) Ligaments
 - (4) Tendons
152. Epithelial tissue can be distinguished from other animal tissues by its:
- (1) large extracellular matrix.
 - (2) contractibility.
 - (3) ability to carry action potentials.
 - (4) ability to act as lining or covering over body cavities.
153. *Periplaneta americana* is:
- (1) dioecious without sexual dimorphism.
 - (2) monoecious without sexual dimorphism.
 - (3) monoecious with sexual dimorphism.
 - (4) dioecious with sexual dimorphism.
154. Which of the following is/are lined by squamous epithelium?
- A. Goblet cells
 - B. Alveoli of lungs
 - C. Dry surface of skin
 - D. Inner lining of ducts of salivary glands
 - E. Endothelium of blood vessels
- Choose the most appropriate answer from the options given below:
- (1) B and E only
 - (2) A, B, C and E only
 - (3) A, D and E only
 - (4) A, B, C and D only
155. Which of the following structures protects the eyes of the frog in water?
- (1) Nictitating membrane
 - (2) Tympanum
 - (3) Bidder's canal
 - (4) Cloaca
156. How many digits does the forelimb and hindlimb of frog have respectively?
- (1) Three and Four
 - (2) Five and Four
 - (3) Four and Five
 - (4) Four and Three

157. Match **List-I** with **List-II** w.r.t cockroach:

List-I		List-II	
(A)	Sclerite	(I)	Forewings
(B)	Tegmina	(II)	10 th segment
(C)	Gizzard	(III)	Exoskeleton plate
(D)	Anal cerci	(IV)	Proventriculus

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
- (2) A-II, B-I, C-III, D-IV
- (3) A-III, B-I, C-IV, D-II
- (4) A-II, B-III, C-I, D-IV

158. Read the following statements (A-E).

- A. Salivary gland is an exocrine gland.
- B. Forewings of cockroaches arise from prothorax.
- C. Frog is a ureotelic animal.
- D. Intercalated disc is present in cardiac muscle.
- E. Frog has a three-chambered heart.

Which of the following statements are **correct**?

- (1) A, B and D only
- (2) D and E only
- (3) B and C only
- (4) A, C, D and E only

159. Vasa efferentia in frogs are _____ in number that arise from testes.

Choose the option which fills the blank **correctly**.

- (1) 10-12
- (2) 12-14
- (3) 15-20
- (4) 20-25

160. All of the following statements are correct, **except**:

- (1) Muscles are mainly of three types.
- (2) Endocrine glands do not have ducts.
- (3) Muscles do not play an active role in all the movements of the body.
- (4) Cuboidal epithelium has cube-like cells.

161. Find the **correct** statement.

- (1) Only male cockroaches bear a pair of short, thread-like anal styles.
- (2) Head of cockroach is formed by the fusion of seven segments.
- (3) The abdomen in both male and female cockroach consists of five segments.
- (4) In cockroaches, only a single thoracic segment bears a pair of legs.

162. Which of the following is **not** a division of hindgut of cockroach?

- (1) Ileum
- (2) Colon
- (3) Duodenum
- (4) Rectum

163. Which of the following type of tissue **correctly** matches with its locations?

- (1) Adipose tissue - Tendons
- (2) Squamous epithelium - Lining of stomach
- (3) Cuboidal epithelium - Tip of nose
- (4) Smooth muscle - Wall of intestine

164. Given below are two statements:

Statement I: In frogs, the forelimbs and hindlimbs help in swimming and burrowing.

Statement II: Body of a frog is divisible into head and trunk.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

165. How many pair of cranial nerves arise from brain of frog?

- (1) 2 pairs
- (2) 3 pairs
- (3) 10 pairs
- (4) 20 pairs

166. Which of the following has a hard and non-pliable ground substance rich in calcium salts and collagen fibres?

- (1) Cartilage
- (2) Bone
- (3) Blood
- (4) Neurons

167. Which of the following is **not** included in the mouthparts of the cockroaches?

- (1) Collateral gland
- (2) Maxilla
- (3) Mandible
- (4) Hypopharynx

168. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Compound epithelium plays major role in absorption and secretion.

Reason R: Compound epithelium is not found in the lining of stomach.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

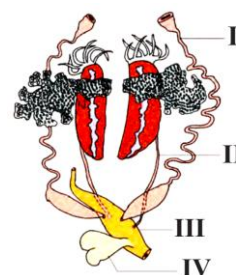
169. Which of the following is **not** a characteristic feature of *Rana tigrina*?

- (1) Webbed feet
- (2) Bilobed tongue
- (3) Thick, leathery skin
- (4) Membranous tympanum

170. Select the **incorrect** statements about the hind wings of cockroach.

- A. They are not used in flying.
 - B. They are also known as mesothoracic wings.
 - C. They are transparent and membranous.
- (1) A only
 - (2) A and B only
 - (3) A and C only
 - (4) A, B and C

171. Refer to the given below diagram.



Which of the following option represent cloaca?

- (1) I
- (2) II
- (3) III
- (4) IV

172. Which of the following statements is **incorrect** w.r.t *Rana tigrina*?

- (1) Frogs have vertebral column.
- (2) Tadpole undergoes metamorphosis to form the adult.
- (3) The hindbrain of frogs is characterised by a pair of optic lobes.
- (4) Brain is enclosed inside cranium.

173. The head of the cockroach shows great mobility in all the directions due to:
- (1) flexible neck.
 - (2) absence of neck.
 - (3) small size of head.
 - (4) presence of compound eyes.

174. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Adipose tissue	(I)	Blood
(B)	Dense irregular connective tissue	(II)	Macrophages
(C)	Fluid connective tissue	(III)	Fat storage
(D)	Areolar tissue	(IV)	Skin

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
 - (2) A-II, B-IV, C-I, D-III
 - (3) A-I, B-II, C-III, D-IV
 - (4) A-III, B-IV, C-I, D-II
175. Which of following option is **correct** w.r.t phallomere in cockroaches?
- (1) It helps to store spermatophores.
 - (2) It is chitinous external genitalia.
 - (3) It is accessory reproductive gland.
 - (4) It represents ejaculatory duct.
176. Choose the **correct** statement w.r.t frog.
- (1) In male frogs, the ureters and oviduct open separately in the cloaca.
 - (2) Frogs do not have renal portal system.
 - (3) The lungs of frogs are a pair of elongated, pink coloured sac-like structures.
 - (4) Sexual dimorphism is absent in frogs.
177. Which of the following statements is **incorrect**?
- (1) Female cockroach possesses sixteen ovarioles in both ovaries.
 - (2) Cockroaches exhibit mosaic vision with less sensitivity and more resolution.
 - (3) A mushroom-shaped gland is present in the 6th-7th abdominal segments of male cockroach.
 - (4) A pair of spermatheca is present in the 6th segment of female cockroach.
178. Select the **incorrect** statement w.r.t cockroach.
- (1) Cockroaches possess open circulating system.
 - (2) Development is paurometabolus.
 - (3) Cockroach are uricotelic animals.
 - (4) Hepatic caeca are 150-200 in number.

179. Which of the following is **correct** pairing of a body part and the kind of muscle tissue that moves it?
- (1) Wall of blood vessel - Smooth muscle fibres
 - (2) Heart wall - Skeletal muscle fibres
 - (3) Biceps - Smooth muscle fibres
 - (4) Intestine - Cardiac muscle fibres

180. Which of the following exerts the greatest control over the body's responsiveness to changing conditions?
- (1) Neural tissue
 - (2) Epithelial tissue
 - (3) Adipose tissue
 - (4) Areolar tissue

181. When two or more organs perform a common function by their physical/chemical interaction, they together form a/an:
- (1) organ.
 - (2) organ system.
 - (3) tissue.
 - (4) organelle.

182. Given below are two statements:
- Statement I:** In unicellular organisms all functions are performed by a single cell.
- Statement II:** Frogs RBCs are nucleated.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.

183. How many among the following are secretions of exocrine glands?

Saliva, earwax, mucus, digestive enzyme, hormone
--

- (1) Four
- (2) Two
- (3) Three
- (4) Five

184. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Frogs show camouflage.

Reason R: Frogs have the ability to change the colour to hide from their enemies.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

185. Which of the following cells support and protect neurons?

- (1) Osteocytes
- (2) Chondrocytes
- (3) RBCs
- (4) Neuroglia

SECTION - B

186. Tendons and ligaments are the examples of:

- (1) loose connective tissue.
- (2) specialised connective tissue.
- (3) dense regular connective tissue.
- (4) dense irregular connective tissue.

187. The functions of bones does **not** involve:

- (1) bearing weight of the body.
- (2) providing protection to soft tissues or organs.
- (3) production of blood cells.
- (4) providing elasticity to body parts.

188. Forebrain of frog includes:

- (1) olfactory lobes.
- (2) medulla oblongata.
- (3) paired diencephalon.
- (4) paired cerebellum.

189. Which of the following tissue covers the moist surface of pharynx?

- (1) Cuboidal epithelium
- (2) Columnar epithelium
- (3) Squamous epithelium
- (4) Compound epithelium

190. The nymphs of cockroaches grow by moulting about _____ times to reach the adult form.

Choose the option which fills the blank **correctly**.

- (1) 6
- (2) 8
- (3) 10
- (4) 13

191. Given below are two statements:

Statement I: Malpighian tubules of cockroach help in removal of excretory products from haemolymph.

Statement II: In female cockroach, 10th sternum is boat shaped.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

192. In frog, oxygen dissolved in water is exchanged through:

- (1) external nares. (2) spiracles.
- (3) buccal cavity. (4) skin.

193. Match **List-I** with **List-II**:

List-I		List-II	
(A)	Cartilage	(I)	Ducts of glands
(B)	Cube-like cells	(II)	Stomach
(C)	Tall and slender cells	(III)	Air sacs of lungs
(D)	Flattened cells	(IV)	Between adjacent bones of limbs

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-III, B-IV, C-II, D-I
- (3) A-II, B-I, C-IV, D-III
- (4) A-IV, B-I, C-II, D-III

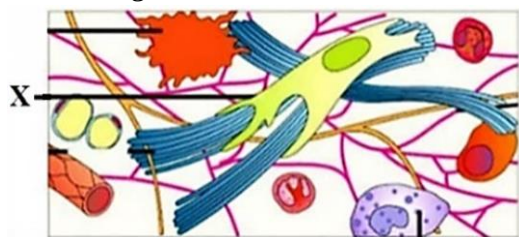
194. Which of the following type of cell junction is responsible for the cementing adjoining cells?

- (1) Tight junctions
- (2) Gap junctions
- (3) Adhering junctions
- (4) All of these

195. Cartilage with solid and pliable matrix that resists compression is present in:

- (1) limbs of adults.
- (2) tip of nose.
- (3) intervertebral disc.
- (4) all of these

196. Refer to the following diagram and identify the tissue along with the structure “X”.



- (1) Adipose tissue, X: Macrophage
(2) Areolar tissue, X: Fibroblast
(3) Adipose tissue, X: Mast cell
(4) Areolar tissue, X: Collagen fibres
197. In cockroaches, the compound eye consists of:
- (1) 200 hexagonal ommatidia.
(2) 20 hexagonal ommatidia.
(3) 2000 hexagonal ommatidia.
(4) 20000 hexagonal ommatidia.
198. Read the following statements (A-E) and choose the **correct** set of statements.
- A. Loose connective tissue has cells and fibre compactly arranged in a semi-fluid ground substance.
B. Tendons attach skeletal muscles to bones.
C. Chondrocytes are enclosed in small cavities within the matrix.
D. Columnar epithelium is made up of single layer of cells.

E. Smooth muscles fibres are fusiform in shape.
Choose the most appropriate answer from the options given below:

- (1) A, C and E only
(2) A, B and D only
(3) B, C and E only
(4) B, C, D and E only
199. How many pairs of spiracles are present on the lateral side of the body of cockroach?
- (1) Five
(2) Six
(3) Ten
(4) Twelve
200. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
- Assertion A:** Frogs help to maintain ecological balance.
Reason R: The frogs serve as an important link in the food chain and food web.
- In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

